

Connectivity in Graphs

Theorem 1

A graph G is bipartite $\iff$ every cycle of G has even length.

Proof. ($\Rightarrow$) Since G is bipartite, we can split its vertex set into two disjoint sets A and B so that each edge of G joins a vertex of A and a vertex of B . Let $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_n \rightarrow v_0$ be a cycle of G , and assume $v_0 \in A$; then $v_1 \in B$, $v_2 \in A$, and so on. Since v_n must lie in B , the cycle has even length.

($\Leftarrow$) Pick any vertex $u \in V(G)$, and define two sets

$$X = \{x \in V : d(u, x) \text{ is even}\}, \quad Y = \{y \in V : d(u, y) \text{ is odd}\},$$

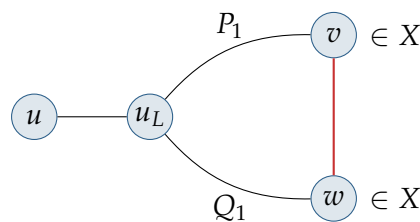
where $d(u, x)$ denotes the length of a shortest path from u to x . We claim (X, Y) is a bipartition, i.e. no edge connects two vertices of X or two vertices of Y .

Suppose, for contradiction, that there exists an edge connecting two vertices in the same set, say $v, w \in X$. Let P be a shortest path from u to v , and Q a shortest path from u to w . Denote by u_L the last common vertex of P and Q (where the two paths diverge), so that

$$P = u \rightarrow \cdots \rightarrow u_L \rightarrow P_1 \rightarrow v, \quad Q = u \rightarrow \cdots \rightarrow u_L \rightarrow Q_1 \rightarrow w.$$

Both P and Q have the same length up to u_L ; since the lengths of P and Q are both even, the lengths of P_1 and Q_1 are also even (because each total is even). Now consider the path $P_1 + \overline{Q_1} + (v, w)$ (i.e. P_1 , followed by Q_1 reversed, followed by the edge (v, w)): this is a cycle of odd length, a contradiction.

The same argument applies to the set Y .



□

Theorem 2

Let G be a simple graph on n vertices. If G has k components, then the number m of edges of G satisfies

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2}.$$

Proof. We prove the lower bound $m \geq n - k$ by induction on the number of edges; the result is trivial if G is a null graph. If G contains as few edges as possible (say m_0) subject to having n vertices and k components, then the removal of any edge of G must increase the number of components by 1, and the remaining graph has n vertices, $k + 1$ components, and $m_0 - 1$ edges. By the induction hypothesis,

$$m_0 - 1 \geq n - (k + 1) \implies m_0 \geq n - k,$$

as required.

To prove the upper bound, we assume that each component of G is complete. Suppose there are two components C_i, C_j with n_i and n_j vertices respectively, $n_i \geq n_j \geq 1$. If we replace C_i and C_j by complete graphs on $n_i + 1$ and $n_j - 1$ vertices, then the total number of edges

$$\frac{(n_i + 1)(n_i)}{2} + \frac{(n_j - 1)(n_j - 2)}{2}$$

is increased by $n_i - n_j + 1 > 0$. It follows that, in order to maximize the number of edges, G must consist of a complete graph on $n - k + 1$ vertices together with $k - 1$ isolated vertices, which gives

$$\frac{(n - k + 1)(n - k)}{2} + 0 \times (k - 1).$$

□

Corollary 3

Any simple graph with n vertices and more than $\frac{(n - 1)(n - 2)}{2}$ edges is connected.

Proof. By Theorem 2,

$$\frac{(n - 1)(n - 2)}{2} \geq \frac{(n - k)(n - k + 1)}{2}, \quad \forall k \geq 2,$$

which implies $k = 1$.

□